EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession.

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. Input:

- The user will enter their **username** and **password** in a **login form**.
- The servlet will validate the credentials and create an **HTTP session** for the user if authentication is successful.

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2. Steps for Implementing Session Tracking with HttpSession:

1. **Create the Login Form (HTML):**
   - Create an HTML form with two input fields for **username** and **password**.
   - Provide a **submit button** to send the data to the servlet for authentication.
2. **Create the Login Servlet:**
   - Write a Java servlet class to handle **login** requests.
   - The servlet will:
     - Retrieve the **username** and **password** from the form.
     - Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded or from a database).
     - Create a new **HTTP session** for the user if the credentials are valid.
     - Set session attributes such as the user's **username** to track the session.
3. **Validate User Credentials:**
   - In the servlet, check if the **username** and **password** match the expected values.
   - If valid, create a session and store the **username** in the session.

- If invalid, redirect the user back to the login page with an error message.

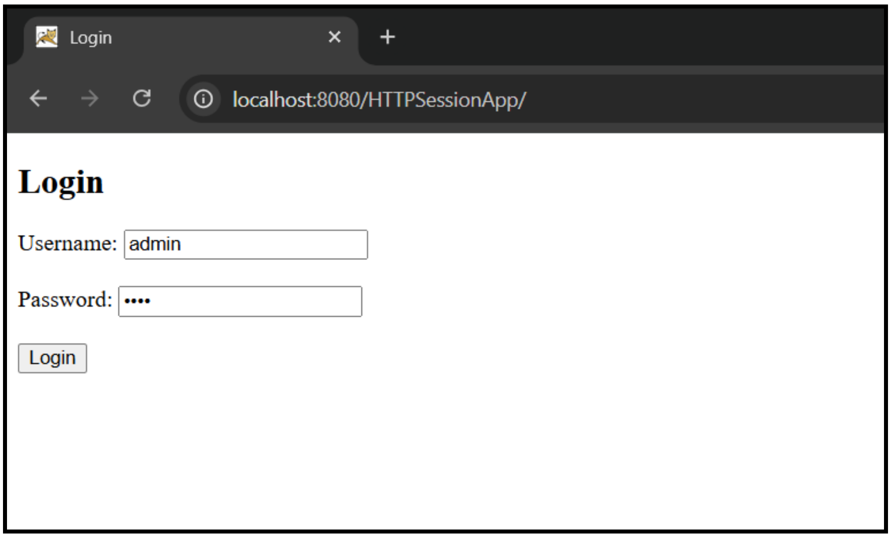
4. **Create and Track User Session:**
    - Upon successful login, use the HttpSession object to store the session data.
    - Use the session.setAttribute("username", username) method to save the **username**.
    - Optionally, set session timeout using session.setMaxInactiveInterval(time_in_seconds) to manage session expiration.

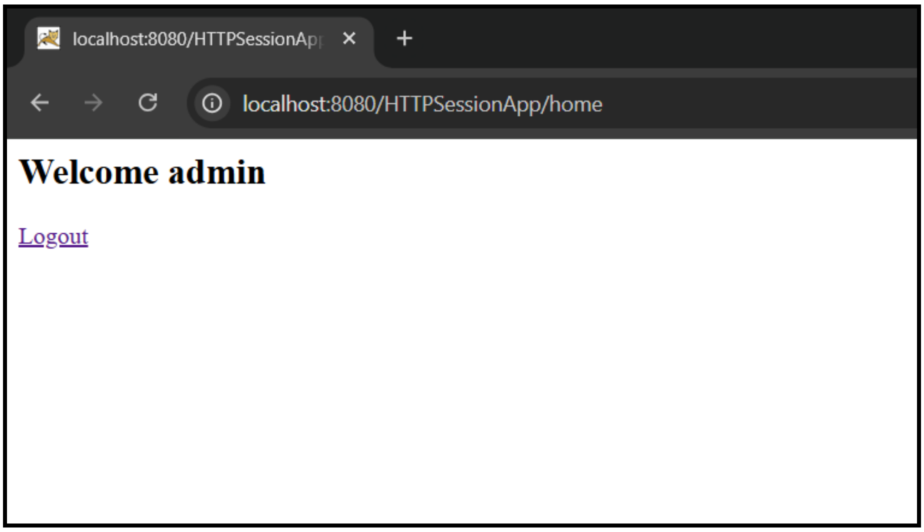
5. **Redirect to a Protected Page:**
    - After a successful login, the user will be redirected to a **protected page** (e.g., a dashboard or home page).
    - The protected page should check if the user is authenticated by checking the **session attribute**.

6. **Logout and Session Invalidation:**
    - Provide a **logout feature** that invalidates the session using session.invalidate() when the user logs out.
    - After logout, redirect the user back to the login page.

7. **Session Expiration:**
    - If the session expires or the user manually logs out, the session will be invalidated, and the user will be redirected to the login page.

Output:





HTML File (index.html):

```
<!DOCTYPE html>
<html>
<head>
    <title>Login</title>
</head>
<body>
    <h2>Login</h2>

    <form action="login" method="post">
        Username: <input type="text" name="username"><br><br>
        Password: <input type="password" name="password"><br><br>
        <input type="submit" value="Login">
    </form>

</body>
</html>
```

Java File (HomeServlet.java):

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class HomeServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        HttpSession session = request.getSession(false);
```

```
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        if (session != null && session.getAttribute("username") != null) {

            String user = (String) session.getAttribute("username");

            out.println("<h2>Welcome " + user + "</h2>");
            out.println("<a href='logout'>Logout</a>");

        } else {
            response.sendRedirect("index.html");
        }
    }
}
```

Java File (LoginServlet.java):

```
import java.io.IOException;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        if ("admin".equals(username) && "1234".equals(password)) {

            HttpSession session = request.getSession();
            session.setAttribute("username", username);

            response.sendRedirect("home");

        } else {
            response.setContentType("text/html");
            response.getWriter().println("<h3>Invalid Login</h3>");
            response.getWriter().println("<a href='index.html'>Try
Again</a>");
        }
    }
}
```

Java File (LogoutServlet.java):

```java
import java.io.IOException;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class LogoutServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if (session != null) {
            session.invalidate();
        }

        response.sendRedirect("index.html");
    }
}
```

File (web.xml):

```xml
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
         version="5.0">

    <!-- Default welcome file -->
    <welcome-file-list>
        <welcome-file>index.html</welcome-file>
    </welcome-file-list>

    <servlet>
        <servlet-name>LoginServlet</servlet-name>
        <servlet-class>LoginServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>LoginServlet</servlet-name>
        <url-pattern>/login</url-pattern>
    </servlet-mapping>

    <servlet>
        <servlet-name>HomeServlet</servlet-name>
        <servlet-class>HomeServlet</servlet-class>
    </servlet>
```

```xml
    <servlet-mapping>
        <servlet-name>HomeServlet</servlet-name>
        <url-pattern>/home</url-pattern>
    </servlet-mapping>

    <servlet>
        <servlet-name>LogoutServlet</servlet-name>
        <servlet-class>LogoutServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>LogoutServlet</servlet-name>
        <url-pattern>/logout</url-pattern>
    </servlet-mapping>

</web-app>
```

RESULT:

A simple login system is created where the user's session is tracked by a servlet to demonstrate session tracking using HttpSession.